MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE FSGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required.

Construction (Watertight)

Sizes 3 to 23 inclusive, extruded silicone rubber insulation

First - Copper conductor, uncoated (see Details for size).
Second - Extruded silicone rubber insulation (see Details for diameter).
Third - Glass braid.
Fourth - Braid covering. Standard Identification Code, applied by method 1.
Fifth - The four conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Either non-fibrous or a combination of non-fibrous and fibrous fillers shall be used to form a firm well-rounded assembly.
Sixth - Binder or combination binder/barrier.
Seventh - Special thermoplastic jacket (gray) (see Details for thickness). Cable surface marking required.

Sizes 50 to 200 inclusive, silicone rubber-glass tape insulation

First - Copper conductor, uncoated (see Details for size).
Second - Silicone rubber-glass tape insulation (see Details for diameter). Letter Identification Code, applied by method 5.
Third - The four conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Either non-fibrous or a combination of non-fibrous and fibrous fillers shall be used to form a firm well-rounded assembly.
Fourth - A binder.
Fifth - Special thermoplastic jacket (gray) (see Details for thickness). Cable surface marking required.

Note 1: When this cable is manufactured with armor, the following additional requirements shall apply:

(D)

(a) The type designation shall change to FSGA.
(b) Standard thermoplastic jacket (black or gray) may be used.
(c) Braided metal armor and paint shall be added.
(d) Cable surface marking will not be required.
(e) The maximum overall dimension shall be increased by 0.050-inch.
(f) Group C inspection shall include: Armor - Conformance to material, construction and coverage. When standard thermoplastic jacket is used, the air oven test does not apply.

(D) denotes changes

FSC 6145

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Details

Type and size	Conductor size Navy standard	Diameter over outer insulating tape nominal (nom) (inch)	Diameter over insulating minimum (min) (inch)	Diameter over braid covering maximum (max) (inch)	Jacket thickness nom (inch)	Overall diameter max (inches)	Conductor resistance per 1000 ft at 25°Celsius (C) max (ohms)	Insulation resistance per 1000 ft min (megohms)	Cold bending mandrel diameter max (inches)	Accelerated service loading (amps)
FSGU-3	3(7)	--	0.096	0.130	0.030	0.447	4.0	500	--	--
FSGU-4	4(7)	--	.112	.143	.040	.513	2.51	500	--	--
FSGU-9	9(7)	--	.154	.187	.040	.630	1.25	500	--	115
FSGU-23	23(7)	--	.284	.316	.050	.950	0.496	500	--	
FSGU-50	50(19)	0.334	--	--	.050	1.050	.23	200	13	185
FSGU-75	75(37)	.407	--	--	.050	1.240	.149	175	15	250
FSGU-100	100(61)	.453	--	--	.060	1.358	.114	160	16	295
FSGU-150	150(61)	.557	--	--	.060	1.625	.0718	135	19	400
FSGU-200	200(61)	.634	--	--	.060	1.820	.0571	125	21	460

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MINATION AND TESTS:

	Requirements
Basic electrical:	
Voltage withstand - Volts, root mean square (Vrms), min	
Sizes 3 through 9	3000
Sizes 23 and larger	5000
Insulation resistance - Megohms/1000 ft, min	(See Details)
Conductor resistance - Ohms/1000 ft, max	(See Details)
Group A:	
Visual and dimensional	
Watertightness - See MIL-C-915, table of limits for water leakage	
Abrasion resistance - Sizes 3 through 23, scrapes, minimum	250
Crack resistance - Sizes 3 through 23.	No damage
Group B:	
Gas flame - 1 hour	No failure
Drip - At 95°C ± 1°C	Zero
Cold bending, cable sizes 50 and larger (See Details for mandrel diameter)	No damage
Physicals (unaged)	
Insulation (extruded)	
Tensile strength - pounds-force per square inch (lbf/in^2), min	750
Elongation - Percent, min	125
Jacket (standard thermoplastic)	
Tensile strength, lbf/in^2, min	1800
Elongation - Percent, min	225
Jacket (special thermoplastic)	
Tensile strength - lbf/in^2, min	2300
Elongation - Percent, min	250
Group C:	
Physicals (aged)	
Jacket (standard thermoplastic)	
Hot oil immersion	
Tensile strength - lbf/in^2, min	1400
Elongation - Percent, min	50
Jacket (special thermoplastic)	
Air oven (168 hours at 136°C)	
Tensile strength - Percent of unaged, min	80
Elongation - Percent of unaged, min	60
Hot oil immersion	
Tensile strength - Percent of unaged, min	75
Elongation - Percent of unaged, min	50
Permanence of printing (conductor) - Sizes 3 through 23, cycles, min	50
Permanence of printing (jacket) - cycles, min	250
Cable filler removability	

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of Group A, B, and C plus the following:

Cable aging and compatibility (125°C ± 5°C)	No failure
Cold working (minus 20°C ± 2°C)	No damage
Gas flame (3 hours)	No failure
Specific gravity of extruded silicone insulation, max	1.55
Accelerated service - Sizes 23 and larger (see Details for load current)	No failure

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UNIT ORDERING LENGTHS:

Type and size	Feet (nom)
FSGU-3	2500
FSGU-4 and -9	2000
FSGU-23 and -50	1500
FSGU-75 through -150	1000
FSGU-200	500

Custodians:
Army - MI
Navy - SH
Air Force - 99

Review activities:
Army - EL, AV
Navy - EC

User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH

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OMB Approval
No. 22-R255

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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